Polynomial equations

1

a $x=\frac{4}{5},-3,2$

b $x=-6,-4,4$

c $x=-3,6,\frac{2}{5}$

d $x=0,\frac{5}{3}$

e $x=5$

f $x=-10,5,0$

g $x=-4,4,0$

h $x=0,\frac{9}{8},-\frac{9}{8}$

2

a $2x^3(x+20)(x-2)=0$

b $x=0,2,-20$

3

a $\left(4x^2-5\right)(x-3)=0$

b $x=3,\frac{\sqrt{5}}{2},-\frac{\sqrt{5}}{2}$

4

a $(x-5)\left(x^2+6x+9\right)=0$

b $x=-3$

Equations reducible to quadratics

5

a

i $p=x^2$

ii $p=9,\ 25$

iii $x=\pm3,\pm5$

b

i $p=x^2$

ii $p=1,\ 25$

iii $x=\pm1,\pm5$

c

i $p=5x^2$

ii $p=4,\ 9$

iii $x=\pm\frac{2\sqrt{5}}{5},\pm\frac{3\sqrt{5}}{5}$

d

i $p=9x^2$

ii $p=4,\ 9$

iii $x=\pm1,\pm\frac{2}{3}$

e

i $p=2x+4$

ii $p=-2,\ -8$

iii $x=-3,-6$

f

i $p=3x+3$

ii $p=-12,\ -9$

iii $x=-5,-4$

g

i $p = 4x + 8$

ii $p = -4, -12$

iii $x = -3, -5$

i $p = 4x - 4$

ii $p = 7,\ -9$

iii $x = \frac{11}{4}, -\frac{5}{4}$

6

a $x = \pm 1, \pm 2$

b $x = \pm 2, \pm 4$

c $x = \pm 5, \pm 2\sqrt{6}$

d $x = \pm 5, \pm 3$

e $x = -4, -\frac{15}{4}$

f $x = -\frac{3}{2}, -\frac{15}{8}$

g $x = -\frac{34}{27}, -\frac{5}{3}$

h $x = \frac{3}{16}, -\frac{1}{4}$

7

a $x = \pm 3$

b $x = \pm 2$

c $x = \pm 4$

d $x = \pm 5$

e $x = \pm\sqrt{7}$

f $x = \pm\sqrt{15}$

g $x = \pm\sqrt{14}$

h $x = \pm 1$

Applications

8

a Any expression equivalent to $4x^3 - 80x^2 + 300x$.

b $x = 4$

9

a $P = x^3 - 7x^2 - 11x + 9$

b $x = 37$

10

a Answers may vary.
Taking out the central rectangular prism leaves behind a triangular prism of length $23x + 10 - 3x$, height $2x$, and width $2x + 6$.
Putting this together, we have

$$\frac{2x\left(23x+10-3x\right)}{2}\left(2x+6\right)=240$$

which we can simplify to

$$x\left(2x+1\right)\left(x+3\right)=12$$

b We can rewrite the equation as

$$2x^3+7x^2+3x-12=0$$

Using long division or otherwize we can factor this into the form:

$$\left(x-1\right)\left(2x^2+9x+12\right)=0$$

The first factor gives the solution $x = 1$, while the second factor has two complex solutions. Since x represents a length (in feet), it must be that $x = 1$.
So the dimensions are as follows:

- The width of the ramp is 8 feet
- The height of the ramp is 2 feet
- The total length of the ramp is 33 feet
- The flat distance between the two sloped sections is 3 feet
- The horizontal length of the shorter sloped section is 12 feet
- The horizontal length of the longer sloped section is 18 feet